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Introduction

- **Problem Statement ID :** 26104
- **Problem Statement Title :** AI-Powered Real-Time Detection and Prevention of Voice Cloning Impersonation Attacks
- **Theme :** Miscellaneous
- **PS Category :** Software
- **Team ID :**
- **Team Name (Registered on Portal) : VoiceOps**



AI-Based Voice Clone Detection and Risk Assessment System

- **Detailed Explanation:**
  A user-friendly app that detects AI-generated voices from uploaded audio and, where live audio access is available, during calls. It provides a real-time risk assessment.
- **How it Addresses the Problem:**
  Reduces manual voice verification, saves time, and alerts users to suspicious voices. Supports pre-call screening and continuous monitoring.
- **Innovation & Uniqueness:**
  Combines AI voice detection, real-time analysis, risk scoring, and instant alerts in one platform. Enables both uploaded-audio analysis and live-call monitoring.



TECHNICAL APPROACH

VOICEOPS

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LIVE CALL AUDIO CAPTURE



Voice Data Collection

- Incoming call audio
- Accept standard audio formats
- Capture speech segments for analysis
- Temporary processing to protect audio data

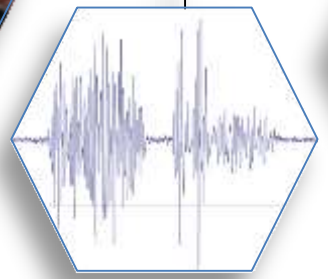
Technology: Python, FastAPI

REAL-TIME AUDIO PREPROCESSING

Clean & Prepare Audio

- Receive short audio chunks from the live call
- Convert audio into a model-compatible representation
- Extract acoustic features such as:
- **MFCC , RMS Energy , Zero-Crossing Rate**

Technology: Librosa + NumPy



AI VOICE DETECTION



Real vs Synthetic Speech

- Analyze speech representations using AI
- Detect patterns associated with cloned/synthetic voices
- Generate an **AI probability score**
- Compare voice characteristics with learned patterns

Technology: Wav2Vec2-based model

Status: *Planned / Integration Stage*

RISK SCORING

Calculate Threat Level

The AI output is converted into an understandable risk level:

Low Risk → Medium Risk → High Risk

AI-generated probability

Output:

Authentic / Suspicious / Potentially Cloned



DATABASE + API



The dashboard displays:

Voice Status

- Genuine
- Suspicious
- Potential Voice Clone

Along with:

AI probability , Detection result , Warning/alert

ALERT & DASHBOARD

Store & Process Results

- Save analysis results
- Provide results to frontend

Technology:

SQLite — database

FastAPI — REST API

Uvicorn — API server

Swagger/OpenAPI — API testing



FEASIBILITY AND VIABILITY

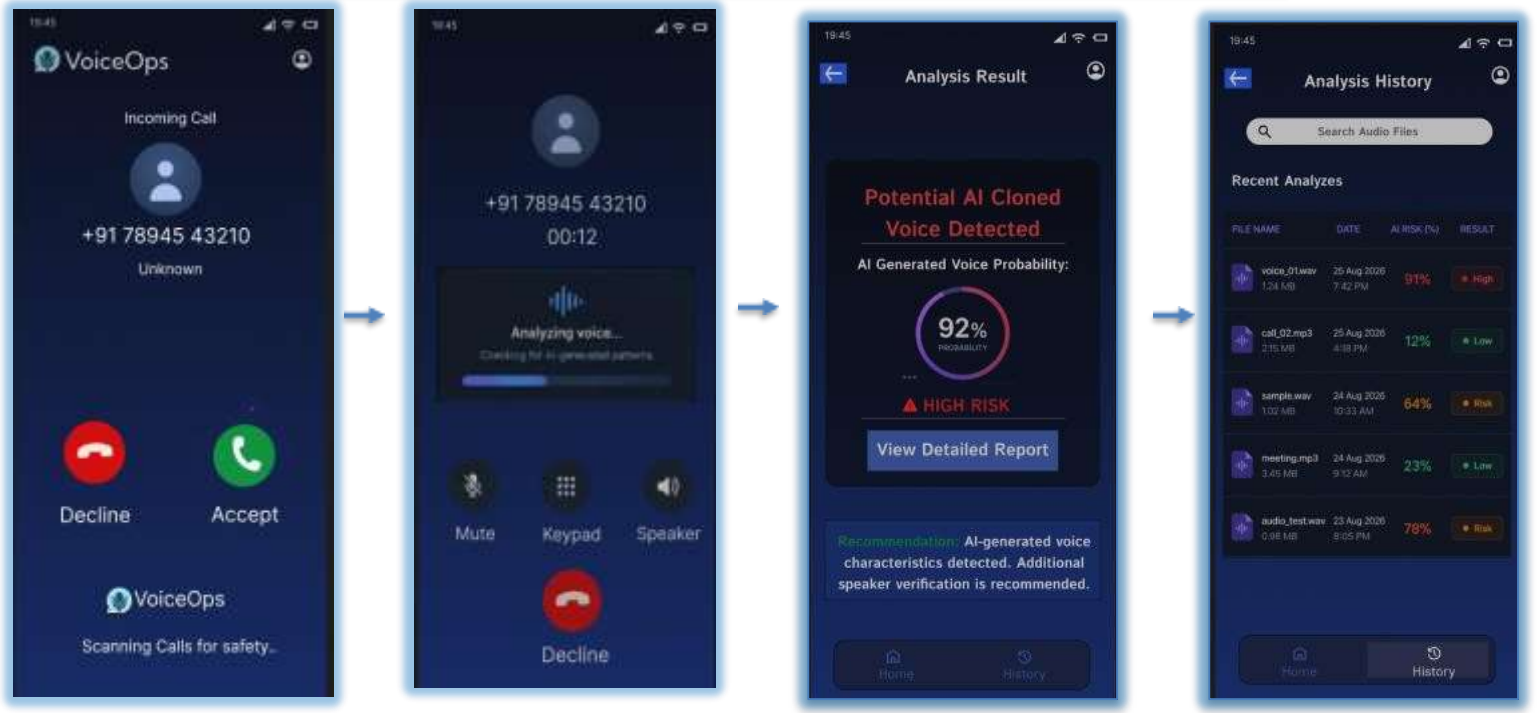
Feasibility

- Uses available technologies Python, FastAPI, Librosa, NumPy and SQLite.
- Wav2Vec2-based AI detection can be integrated with the trained model.
- Figma prototype shows a practical user flow from voice input to detection and alert.

Potential Challenges

- Detecting high-quality AI-generated voices accurately.
- Background noise and poor audio quality may affect results.
- Real-time detection requires processing latency.
- Voice data requires privacy and secure handling.
- Mitigation Strategies

FIGMA PROTOTYPE – USER JOURNEY



Figma link - SIH Voice Cloning App - Figma

VOICEOPS

IMPACT AND BENEFITS

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Impact

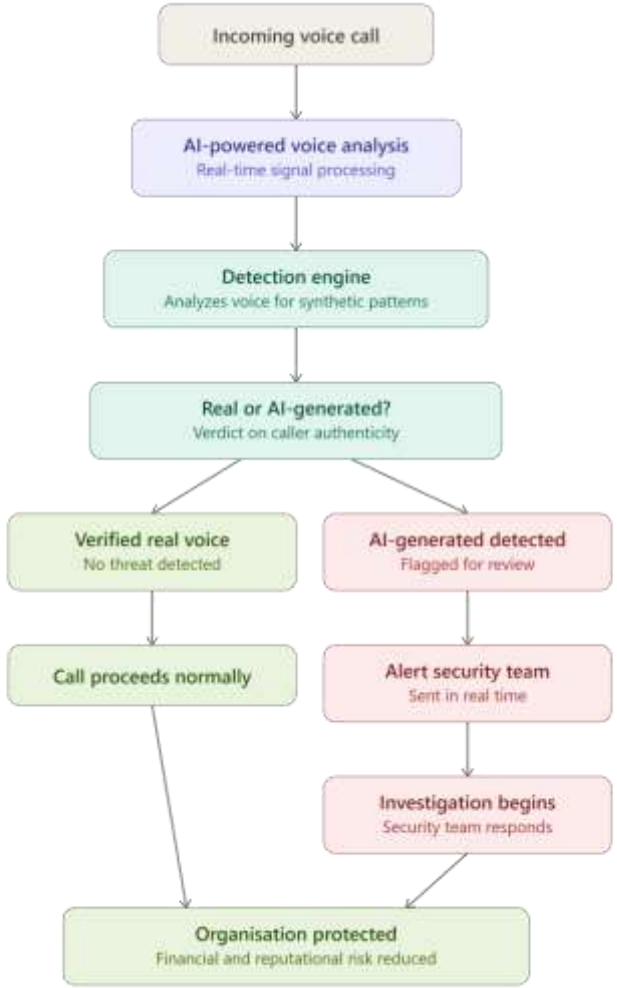
- Prevents AI-based fake voice scams
- Protects organisations and sensitive information
- Detects suspicious voice calls quickly

- Verifies whether a voice is real or AI-generated
- Reduces financial and reputation losses
- Supports security teams in investigations

Benefits

- Real-time voice detection
- AI-powered voice analysis
- Early warning against voice attacks

- Protection of confidential data
- Easy integration with existing systems
- Scalable across different Organisations



VOICEOPS

RESEARCH AND REFERENCES

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Key Research

- ASVspoof 2021 – DeepFake (DF): Benchmark for detecting bonafide and spoofed/deepfake speech.
- Wav2Vec 2.0: Self-supervised speech representation model that can provide speech features for AI-based analysis.
- Audio Deepfake Detection Survey: Reviews current detection methods, datasets, features and challenges in audio deepfake detection.
- Librosa: Python library used for audio processing and feature extraction.
- Research Insight
- Key Gap: Current detection methods can struggle with unknown voice-cloning techniques, noisy audio and generalization to real-world recordings.